

ITN — Packet Tracer Activity

VLAN Configuration

Switches: S1, S2, S3

1. Addressing Table

The topology consists of three switches (S1, S2, S3) providing access-layer connectivity for six PCs across three VLANs. The addressing table below summarises each PC's IP configuration and VLAN membership:

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	172.17.10.21	255.255.255.0	10
PC2	NIC	172.17.20.22	255.255.255.0	20
PC3	NIC	172.17.30.23	255.255.255.0	30
PC4	NIC	172.17.10.24	255.255.255.0	10
PC5	NIC	172.17.20.25	255.255.255.0	20
PC6	NIC	172.17.30.26	255.255.255.0	30

2. Part 1 — View the Default VLAN Configuration

2.1 Display the Current VLANs

On S1, the following command was issued to display all VLANs configured on the switch:

```
S1# show vlan brief
```

By default, every interface (Fa0/1 – Fa0/24, Gig0/1 – Gig0/2) is assigned to VLAN 1, confirming that no VLAN segmentation has been applied yet.

The screenshot shows the Cisco Packet Tracer interface. On the left, the 'Logical' view displays a list of VLANs configured on switch S1:

- VLAN 10: Faculty/Staff
- VLAN 20: Students
- VLAN 30: Guest(Default)
- VLAN 99: Management&Native

On the right, the CLI window shows the output of the command `show vlan brief`:

```

VLAN Name                Status    Ports
-----
1    default                 active    Fa0/1, Fa0/2, Fa0/3, Fa0/4,
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8,
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12,
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
100  fa0-100/24               active    Fa0/100, Fa0/101, Fa0/102, Fa0/103,
                                           Fa0/104, Fa0/105, Fa0/106, Fa0/107,
                                           Fa0/108, Fa0/109, Fa0/110, Fa0/111,
                                           Fa0/112, Fa0/113, Fa0/114, Fa0/115,
                                           Fa0/116, Fa0/117, Fa0/118, Fa0/119,
                                           Fa0/120
101  fa0-101/24               active    Fa0/101, Fa0/102, Fa0/103, Fa0/104,
                                           Fa0/105, Fa0/106, Fa0/107, Fa0/108,
                                           Fa0/109, Fa0/110, Fa0/111, Fa0/112,
                                           Fa0/113, Fa0/114, Fa0/115, Fa0/116,
                                           Fa0/117, Fa0/118, Fa0/119, Fa0/120
102  fa0-102/24               active    Fa0/102, Fa0/103, Fa0/104, Fa0/105,
                                           Fa0/106, Fa0/107, Fa0/108, Fa0/109,
                                           Fa0/110, Fa0/111, Fa0/112, Fa0/113,
                                           Fa0/114, Fa0/115, Fa0/116, Fa0/117,
                                           Fa0/118, Fa0/119, Fa0/120
103  fa0-103/24               active    Fa0/103, Fa0/104, Fa0/105, Fa0/106,
                                           Fa0/107, Fa0/108, Fa0/109, Fa0/110,
                                           Fa0/111, Fa0/112, Fa0/113, Fa0/114,
                                           Fa0/115, Fa0/116, Fa0/117, Fa0/118,
                                           Fa0/119, Fa0/120
104  fa0-104/24               active    Fa0/104, Fa0/105, Fa0/106, Fa0/107,
                                           Fa0/108, Fa0/109, Fa0/110, Fa0/111,
                                           Fa0/112, Fa0/113, Fa0/114, Fa0/115,
                                           Fa0/116, Fa0/117, Fa0/118, Fa0/119,
                                           Fa0/120

```

Screenshot: show vlan brief on S1 — all ports in default VLAN 1

2.2 Verify Connectivity Between PCs on the Same Network

Ping tests were performed between PCs sharing the same subnet. Since all ports remain in VLAN 1, connectivity succeeded in every case:

- PC1 can ping PC4
- PC2 can ping PC5
- PC3 can ping PC6

Pings to hosts on other networks (different subnets) fail, as no Layer 3 routing has been configured between them.

Question: What benefits can VLANs provide to the network?

- Broadcast domain segmentation — broadcast traffic is contained within its own VLAN, reducing unnecessary congestion on the rest of the network.
- Improved security — sensitive groups (e.g., Faculty/Staff, Management) can be logically isolated from other users; inter-VLAN traffic must pass through a Layer 3 device, where it can be filtered.
- Simplified network management — users can be grouped by function or department rather than physical location, allowing devices to be moved or added without re-cabling.
- Reduced cost — a single physical switch can serve multiple logical networks, removing the need for separate switches per group.
- Better performance — limiting broadcast and unnecessary traffic to relevant segments improves overall network efficiency.

3. Part 2 — Configure VLANs

3.1 Create and Name VLANs on S1

The following VLANs were created and named on S1. VLAN names are case-sensitive and must match the required naming exactly:

VLAN ID	Name
10	Faculty/Staff
20	Students
30	Guest(Default)
99	Management&Native
150	VOICE

```

S1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 10
S1(config-vlan)#name Faculty/Staff
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#naame student
^
% Invalid input detected at '^' marker.

S1(config-vlan)#name student
S1(config-vlan)#
S1(config-vlan)#name Student
S1(config-vlan)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   Faculty/Staff          active
20   Student                active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
S1#
S1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 30
S1(config-vlan)#name Guest(Default)
S1(config-vlan)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vlan 20
S1(config-vlan)#name Students
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 99
S1(config-vlan)#name Management&Native
S1(config-vlan)#exit
S1(config)#vlan 150
S1(config-vlan)#name VOICE
S1(config-vlan)#

```

Screenshot: VLAN creation on S1, including correction of a naming typo

Final verification on S1 after all VLANs were created:

```

S1#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Faculty/Staff           active
20   Students                active
30   Guest(Default)          active
99   Management&Native       active
150  VOICE                    active
1002 fddi-default            active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
S1#

```

Screenshot: show vlan brief on S1 — all five VLANs created and named

3.2 Verify the VLAN Configuration

Question: Which command will only display the VLAN name, status, and associated ports on a switch?

Answer: show vlan brief — this command displays only the VLAN ID, Name, Status, and Ports in a condensed summary. It differs from show vlan id [number] or show vlan name [name], which display detail for a single VLAN, and from show vlan summary, which omits names and ports entirely.

3.3 Create the VLANs on S2 and S3

The same commands from Step 1 were used to create and name the identical VLANs on S2 and S3, ensuring consistent VLAN numbering and naming across the topology.

S2:

```

S2>
S2>en
S2#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#vlan 10
S2(config-vlan)#name Faculty/Staff
S2(config-vlan)#exit
S2(config)#vlan 20
S2(config-vlan)#name Students
S2(config-vlan)#exit
S2(config)#vlan 30
S2(config-vlan)#name Guest(Default)
S2(config-vlan)#exit
S2(config)#vlan 99
S2(config-vlan)#name Management&Native
S2(config-vlan)#exit
S2(config)#vlan 150
S2(config-vlan)#name VOICE
S2(config-vlan)#exit
S2(config)#

```

Screenshot: VLAN creation and naming on S2

S3:

```

S3>
S3>en
S3#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#vlan 10
S3(config-vlan)#name Favulty/Staff
S3(config-vlan)#name Faculty/Staff
S3(config-vlan)#exit
S3(config)#vlan 20
S3(config-vlan)#name Students
S3(config-vlan)#exit
S3(config)#vlan 30
S3(config-vlan)#name Guest(Default)
S3(config-vlan)#exit
S3(config)#vlan 99
S3(config-vlan)#name Management&Native
S3(config-vlan)#exit
S3(config)#vlan 150
S3(config-vlan)#name VOICE
S3(config-vlan)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console
S3#

```

Screenshot: VLAN creation on S3, including correction of a naming typo

3.4 Verify the VLAN Configuration

show vlan brief was run on both S2 and S3 to confirm that all five VLANs were created with the correct names on each switch (no ports assigned yet):

```

S2#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Faculty/Staff          active
20   Students              active
30   Guest(Default)        active
99   Management&Native      active
150  VOICE                  active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
S2#
S2#

```

Screenshot: show vlan brief on S2

```

S3#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Faculty/Staff           active
20   Students                 active
30   Guest(Default)           active
99   Management&Native        active
150  VOICE                     active
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default         active
1005 trnet-default           active
S3#
S3#

```

Screenshot: show vlan brief on S3

4. Part 3 — Assign VLANs to Ports

4.1 Assign VLANs to the Active Ports on S2

Interfaces were configured as access ports and assigned to their respective VLANs as follows:

VLAN	S2 Interface	S3 Interface
10	FastEthernet0/11	FastEthernet0/11 (+ Voice VLAN 150)
20	FastEthernet0/18	FastEthernet0/18
30	FastEthernet0/6	FastEthernet0/6

```

S2#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#int f0/11
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 10
S2(config-if)#exit
S2(config)#int f0/18
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 20
S2(config-if)#exit
S2(config)#int f0/6
S2(config-if)#switchport mode access
S2(config-if)#switchport access vlan 30
S2(config-if)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console
S2#

```

Screenshot: access VLAN assignment on S2

Verification on S2 confirms each port is now assigned to the correct VLAN:

```
S2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Faculty/Staff	active	Fa0/11
20	Students	active	Fa0/18
30	Guest (Default)	active	Fa0/6
99	Management&Native	active	
150	VOICE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S2#
```

Screenshot: show vlan brief on S2 after port assignment

4.2 Assign VLANs to the Active Ports on S3

S3 uses the same access port assignment as S2 (VLAN 10: Fa0/11, VLAN 20: Fa0/18, VLAN 30: Fa0/6):

```
S3#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#int f0/11
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 10
S3(config-if)#exit
S3(config)#int f0/18
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 20
S3(config-if)#exit
S3(config)#int f0/6
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 30
S3(config-if)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console
```

Screenshot: access VLAN assignment on S3

4.3 Assign the VOICE VLAN to FastEthernet 0/11 on S3

F0/11 on S3 connects simultaneously to a Cisco IP Phone and PC4. This interface must therefore support user traffic on VLAN 10 and voice traffic on VLAN 150, with QoS enabled to trust the Class of Service (CoS) values marked by the phone:

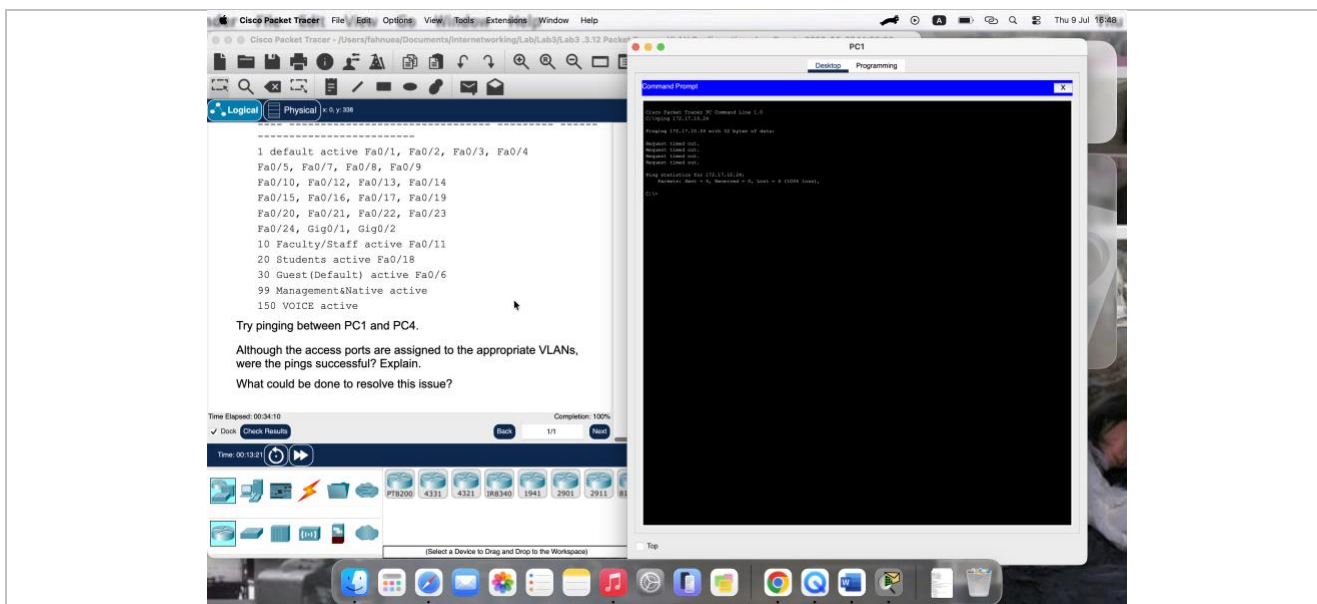
```
S3(config)# interface f0/11
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
```

```
S3#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#int f0/11
S3(config-if)#mls qos trust cos
S3(config-if)#switchport voice vlan 150
S3(config-if)#
```

Screenshot: Voice VLAN and QoS trust CoS configured on S3 F0/11

4.4 Verify Loss of Connectivity

Previously, PCs sharing the same subnet could ping each other successfully while all ports remained in VLAN 1. After assigning access ports to VLAN 10, PC1 (172.17.10.21) was tested against PC4 (172.17.10.24):



Screenshot: ping from PC1 to PC4 — Request timed out, 100% loss

Question: Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

Answer: No — the pings failed (Request timed out, 100% loss) even though PC1 and PC4 are both members of VLAN 10 on the same subnet. This is because the inter-switch links (S1–S2 and S2–S3, via Gig0/1 and Gig0/2) have not been configured as trunk ports. These links remain standard access ports in VLAN 1, so traffic tagged for VLAN 10 on S1 has no path to reach S3, where PC4 is connected.

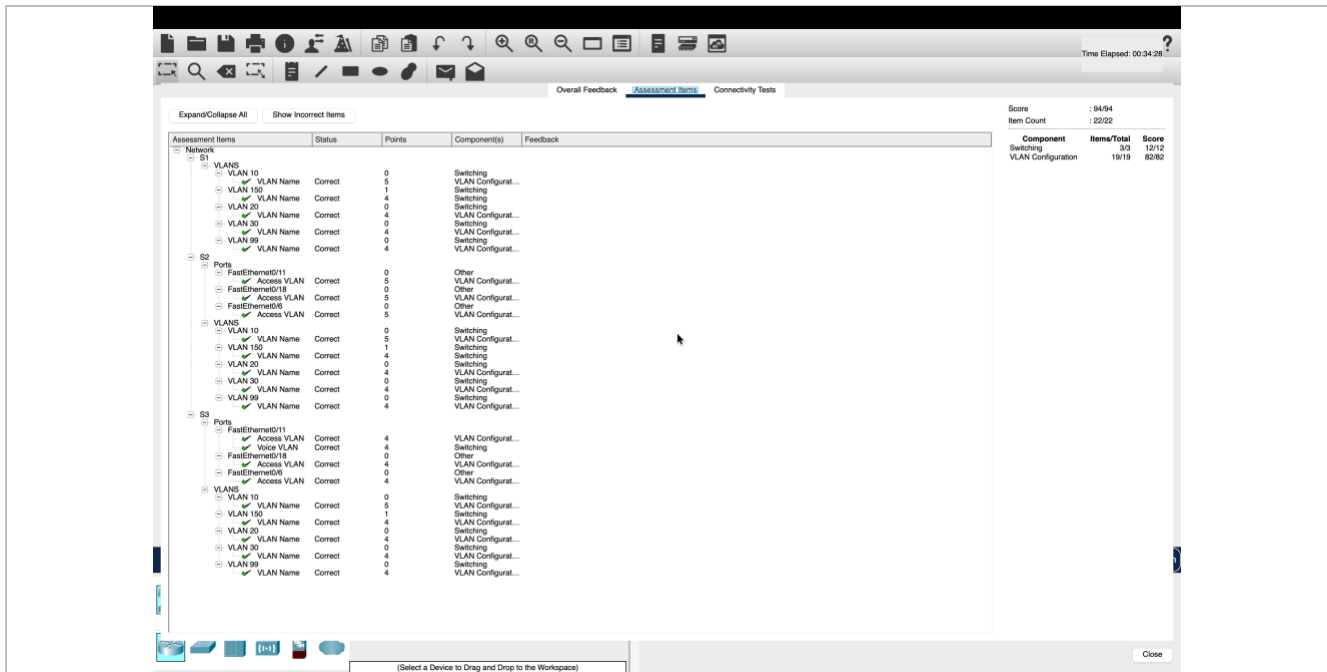
Question: What could be done to resolve this issue?

Answer: The inter-switch links must be configured as trunk ports using switchport mode trunk, with a matching native VLAN (VLAN 99, as configured earlier) on both ends. This allows traffic for all VLANs — including VLAN 10 — to be carried correctly between S1, S2, and S3.

5. Assessment Result

The Check Results feature in Packet Tracer confirmed a final score of 94/94 (100%), with all 22 assessment items completed successfully across VLAN configuration and switching/port assignment on S1, S2, and S3:

Component	Items / Total	Score
Switching	3 / 3	12 / 12
VLAN Configuration	19 / 19	82 / 82
Total	22 / 22	94 / 94



Screenshot: Assessment Items — final score 94/94 (100%)

6. Additional Technical Information

6.1 Why VLAN Names Must Match Exactly

VLAN names configured with the name command are case-sensitive and stored verbatim in the VLAN database (vlan.dat). Any deviation — including capitalisation, spacing, or special characters such as "/" and "&" — is treated as a different value and will fail automated verification, even though the VLAN ID itself may be correct.

6.2 Access Ports vs. Trunk Ports

Attribute	Access Port	Trunk Port
Purpose	Connects a single end device to one VLAN	Carries traffic for multiple VLANs between switches or to a router
Tagging	Untagged — device is unaware of VLANs	Tagged (802.1Q), except for the native VLAN
Typical use	PC, printer, IP phone data port	Switch-to-switch or switch-to-router uplink

6.3 Purpose of the Voice VLAN Feature

The switchport voice vlan command allows a single access port to carry two traffic streams: untagged data traffic in the configured access VLAN, and tagged voice traffic in the specified voice VLAN. This lets a PC and an IP phone share one physical cable and switch port while keeping their traffic logically separated, which is essential for applying QoS priority to voice packets.

6.4 Why Voice Traffic Requires mls qos trust cos

IP phones mark their voice frames with a Class of Service (CoS) value to request priority handling. By default, a switch does not trust CoS markings from an endpoint. The mls qos trust cos command instructs the switchport to trust and honour the CoS values received from the phone, allowing voice traffic to be prioritised end-to-end and helping maintain acceptable call quality.